THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 13: **Our Environment** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the herbivores are removed from a food chain, the secondary consumers 1

(a) become producers **(b)** increase in number

(c) are not affected **(d)** lose their food source

2. Assertion (A): Green plants are producers. Reason (R): They convert solar energy into chemical energy. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

3. A pond contains DDT at 0.003 ppm in water, 0.04 ppm in plankton, 2 ppm in fish and 25 ppm in fish-eating birds. This shows 1

(a) biomagnification **(b)** photosynthesis **(c)** decomposition **(d)** eutrophication

4. A heavy rainfall washes pesticides from a farm into a pond. Within a food chain in the pond, the organism with the highest concentration is 1

(a) small fish **(b)** a fish-eating bird

(c) phytoplankton **(d)** zooplankton

5. Assertion (A): A food web is more stable than a food chain. Reason (R): In a food web many alternative paths are available for the flow of energy. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

6. Which statement about decomposers is correct? 1

(a) They are consumers of living plants

(b) They can act at every trophic level on dead matter

(c) They are producers

(d) They are at only the last level

7. The reason that non-biodegradable substances are harmful is that they 1

(a) release oxygen **(b)** persist and accumulate

(c) increase soil fertility **(d)** decompose too quickly

8. Assertion (A): Herbivores have less energy than producers. Reason (R): Only 10 per cent of the energy is transferred from producers to herbivores. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

9. **Assertion (A):** Harmful chemicals have the highest concentration in the top consumers of a food chain. 1
Reason (R): These chemicals are non-biodegradable and accumulate at each successive trophic level.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Biodegradable substances are broken down by microorganisms. 1
Reason (R): Non-biodegradable substances such as plastics persist in the environment for a very long time.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do we say that pesticides sprayed on crops affect organisms far away? 2

12. How does the loss of ozone in Antarctica affect marine ecosystems there? 2

13. Why does the amount of usable energy decrease at each higher trophic level? 2

14. Give two reasons why it is advisable to use fewer aerosol sprays. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Give reasons: 3
(a) a food web is stable
(b) a pesticide such as DDT is banned
(c) the energy of the Sun is needed daily.

16. Explain how an increase in the human population may affect trophic levels. 3

17. Why is the removal of green cover from a hill-slope harmful to the ecosystem? 3

OR

Why does the ozone hole form mainly over polar regions? 3

18. How can you reduce the use of plastic at home? Mention three steps and the benefit of each. 3

OR

Explain why dumping untreated sewage into a pond may eventually destroy the pond ecosystem. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Scientists observed in 1980s that the amount of ozone in the upper atmosphere over Antarctica fell sharply, creating a hole. Chlorofluorocarbons (CFCs) used in refrigerators and aerosol sprays were identified as the cause. In 1987 the United Nations Environment Programme (UNEP) succeeded in forging an agreement to freeze CFC production. 5
(i) Write the molecular formula of ozone.
(ii) How is ozone formed in the upper atmosphere?
(iii) What is the function of the ozone layer?
(iv) State two harmful effects of ozone depletion on human beings.

20. Evaluate the claim: 'If all the carnivores in a forest are killed, the herbivores will flourish forever.' Use the food web idea. 5

OR

A forest ecosystem has trees, deer, rabbits, tigers, snakes, fungi and bacteria. Classify the organisms by role, write two food chains from the list, and explain what would happen if the fungi and bacteria disappeared. 5

END OF QUESTION PAPER — Total: 40 Marks